

Week 2

Scientific Question Formulation & AI-Assisted Literature Mining

From Curiosity to a Verified, Data-Driven Question

Bioinformatics: From Multi-Omics Data to Discovery

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Two 40-Minute Sessions | Fall 2026

b Statistical modelling between features

Features can be matched between modalities by modelling a statistical relationship

c Latent space inference

Mapping

Latent model prediction

Latent space

Mapping

Latent model prediction

d Consensus of individual inferences (late integration)

DeepSeek Harness installation and NVIDIA API

管理员 PowerShell 执行 Set-ExecutionPolicy RemoteSigned -Scope CurrentUser

```
Windows PowerShell
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PS C:\WINDOWS\system32> $env:DSH_HOME="D:\DeepSeek harness\dsh_data"; npx -y @deepseek-ai/dsh web
npx : 无法加载文件 D:\harness\npx.ps1，因为在此系统上禁止运行脚本。有关详细信息，请参阅
https://go.microsoft.com/fwlink/?LinkID=135170 中的 about_Execution_Policies。
所在位置 行:1 字符: 47
+ ~~~~~ $env:DSH_HOME="D:\DeepSeek harness\dsh_data"; npx -y @deepseek-ai/dsh ...
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess
PS C:\WINDOWS\system32> Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
Set-ExecutionPolicy: 操作成功。

执行策略更改
执行策略可帮助你防止执行不信任的脚本。更改执行策略可能会产生安全风险，如 https://go.microsoft.com/fwlink/?LinkID=135170
中的 about_Execution_Policies 帮助主题所述。是否要更改执行策略？
[Y] 是(Y) [A] 全是(A) [N] 否(N) [L] 全否(L) [S] 暂停(S) [?] 帮助 (默认为“N”)： _

Added to clipboard
```

输入Y，回车

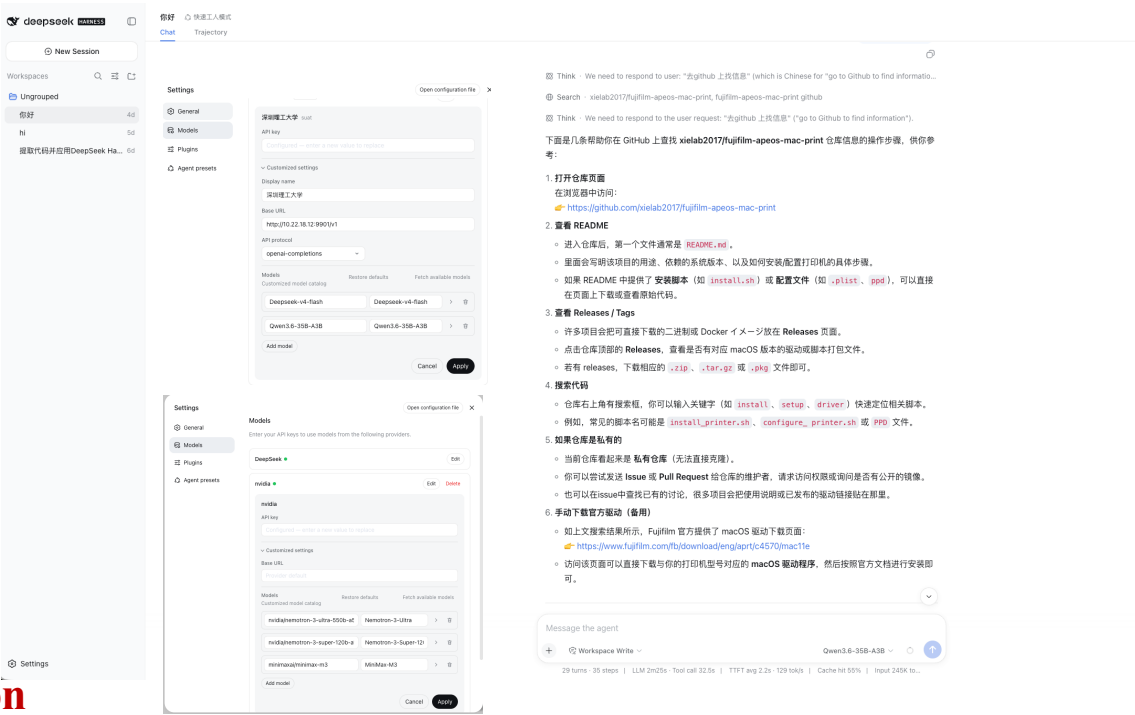
运行\$env:DSH_HOME="D:\DeepSeek harness\dsh_data"; npx -y @deepseek-ai/dsh web
光标闪烁，等待安装，时间较长

```
npm exec @deepseek-ai/dsh web
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> $env:DSH_HOME="D:\DeepSeek harness\dsh_data"; npx -y @deepseek-ai/dsh web
npx : 无法加载文件 D:\harness\npx.ps1，因为在此系统上禁止运行脚本。有关详细信息，请参阅
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中的 about_Execution_Policies 帮助主题所述。是否要更改执行策略？
[Y] 是(Y) [A] 全是(A) [N] 否(N) [L] 全否(L) [S] 暂停(S) [?] 帮助 (默认为“N”)： y
PS C:\WINDOWS\system32> $env:DSH_HOME="D:\DeepSeek harness\dsh_data"; npx -y @deepseek-ai/dsh web
=
```

Give it 40-50 mins to finish the installation



DeepSeek Harness Interface

deepseek HARNESS

你好 快速工人模式

Chat Trajectory

New Session

Workspaces

Ungrouped

你好 4d

hi 5d

提取代码并应用DeepSeek Ha... 6d

Set different folder for different project if you have files input and output

Think · We need to respond to user: "去github 上找信息" (which is Chinese for "go to Github to find information...")

Search · xielab2017/fujifilm-apeos-mac-print, fujifilm-apeos-mac-print github

Think · We need to respond to the user request: "去github 上找信息" ("go to Github to find information").

下面是几条帮助你在 GitHub 上查找 xielab2017/fujifilm-apeos-mac-print 仓库信息的操作步骤, 供你参考:

1. 打开仓库页面
在浏览器中访问:
<https://github.com/xielab2017/fujifilm-apeos-mac-print>
2. 查看 README
 - 进入仓库后, 第一个文件通常是 `README.md`。
 - 里面会写明该项目的用途、依赖的系统版本、以及如何安装/配置打印机的具体步骤。
 - 如果 README 中提供了 `安装脚本` (如 `install.sh`) 或 `配置文件` (如 `.plist`、`ppd`), 可以直接在页面上下载或查看原始代码。
3. 查看 Releases / Tags
 - 许多项目会把可直接下载的二进制或 Docker 镜像放在 `Releases` 页面。
 - 点击仓库顶部的 `Releases`, 查看是否有对应 macOS 版本的驱动或脚本打包文件。
 - 若有 releases, 下载相应的 `.zip`、`.tar.gz` 或 `.pkg` 文件即可。
4. 搜索代码
 - 仓库右上角有搜索框, 你可以输入关键字 (如 `install`、`setup`、`driver`) 快速定位相关脚本。
 - 例如, 常见的脚本名可能是 `install_printer.sh`、`configure_printer.sh` 或 `PPD` 文件。
5. 如果仓库是私有的
 - 当前仓库看起来是 `私有仓库` (无法直接克隆)。
 - 你可以尝试发送 `Issue` 或 `Pull Request` 给仓库的维护者, 请求访问权限或询问是否有公开的镜像。
 - 也可以在issue中查找已有的讨论, 很多项目会把使用说明或已发布的驱动链接贴在那里。
6. 手动下载官方驱动 (备用)
 - 如上文搜索结果所示, Fujifilm 官方提供了 macOS 驱动下载页面:
<https://www.fujifilm.com/fb/download/eng/aprt/c4570/mac1e>
 - 访问该页面可以直接下载与你的打印机型号对应的 `macOS 驱动程序`, 然后按照官方文档进行安装即可。

Read Only

Workspace Write ✓

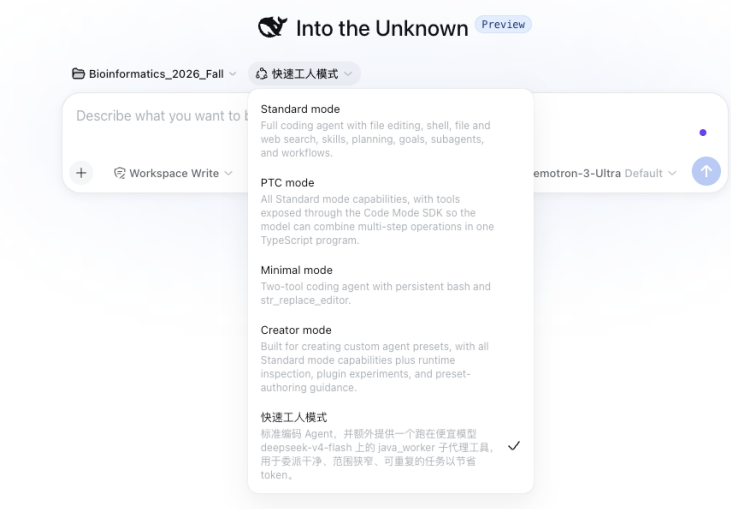
Full access

Qwen3.6-35B-A3B

29 turns · 35 steps | LLM 2m25s · Tool call 32.5s | TTFT avg 2.2s · 129 tok/s | Cache hit 55% | Input 245K to...

Some jobs require long memory and more tokens to process. If you hit the maximum, just ask the agent to continue. The job will keep going till the end.

DeepSeek Harness Interface



Context injection · user-approval

安装这个skill 应用于生物信息课程学习, <https://github.com/wp-a/nature-academic-search>

16:08

1. The DSH (DeepSeek Harness) uses a separate Bundle `dsh-academic-paper-search`
2. It's installed via: `dsh plugin --profile web add dsh-academic-paper-search`
3. Or from source: clone the bundle repository and add it locally

Let me check the source structure for the bundle. Looking at the README, there's a separate repository <https://github.com/wp-a/dsh-academic-paper-search> that needs to be installed. Let me clone that and install it.

- Bash · Clone the DSH bundle repository
- Bash · Check contents of DSH bundle repository
- Read · </tmp/dsh-academic-paper-search/README.md>
- Read · </tmp/dsh-academic-paper-search/package.json>
- Read · </tmp/dsh-academic-paper-search/index.js>
- Read · </tmp/dsh-academic-paper-search/cordis.patch.yml>

Now I have a complete understanding. Let me check if DSH is installed and try to install the bundle.

- Bash · Check if DSH is installed and its version

DSH is installed. Let me also check if `uvx` and `pnpm` are available.

- Bash · Check required dependencies for the bundle
- Bash · Check available tools individually
- Bash · Check if pnpm is available

Good - `uvx` is available, `node` is installed. `pnpm` is not installed. Let me try to install the bundle from source. First, let me check the DSH plugin command syntax.

- Bash · Check DSH plugin command help
- Bash · Check DSH plugin command with profile
- Bash · Check DSH options with profile

`pnpm` is required. Let me install pnpm via npm.

- Bash · Install pnpm globally
- Bash · Verify pnpm is installed

Deep diving... 28m 55s

Week 2 moves students from curiosity to a verified analysis plan

Learning Architecture

Curiosity

Question

Search

Evidence

Gap

Plan



Week 2 moves students from curiosity to a verified analysis plan

Learning Architecture



Three Core Habits



Ask Before Analyzing

Formulate a precise question before touching any dataset



Search Before Claiming

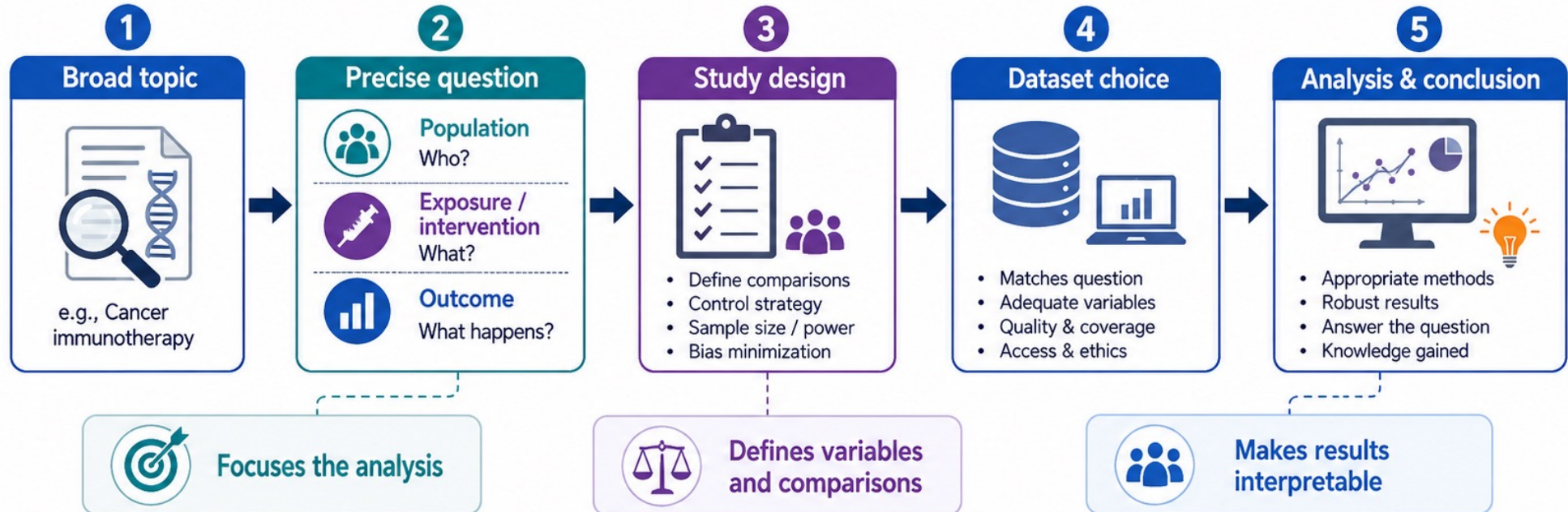
Build reproducible PubMed queries before making assertions



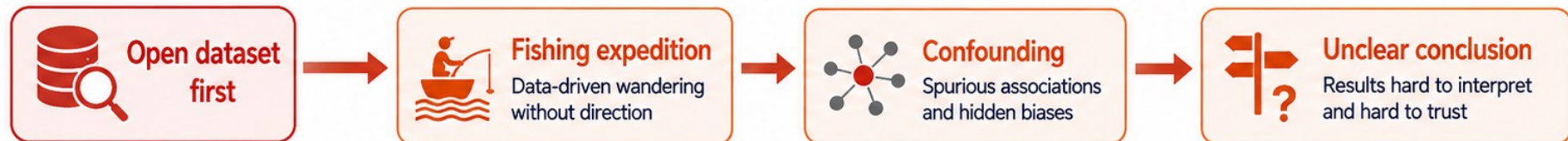
Verify Before Citing

Check every AI-generated claim against real sources

1. Why formulate a precise question before touching any dataset?



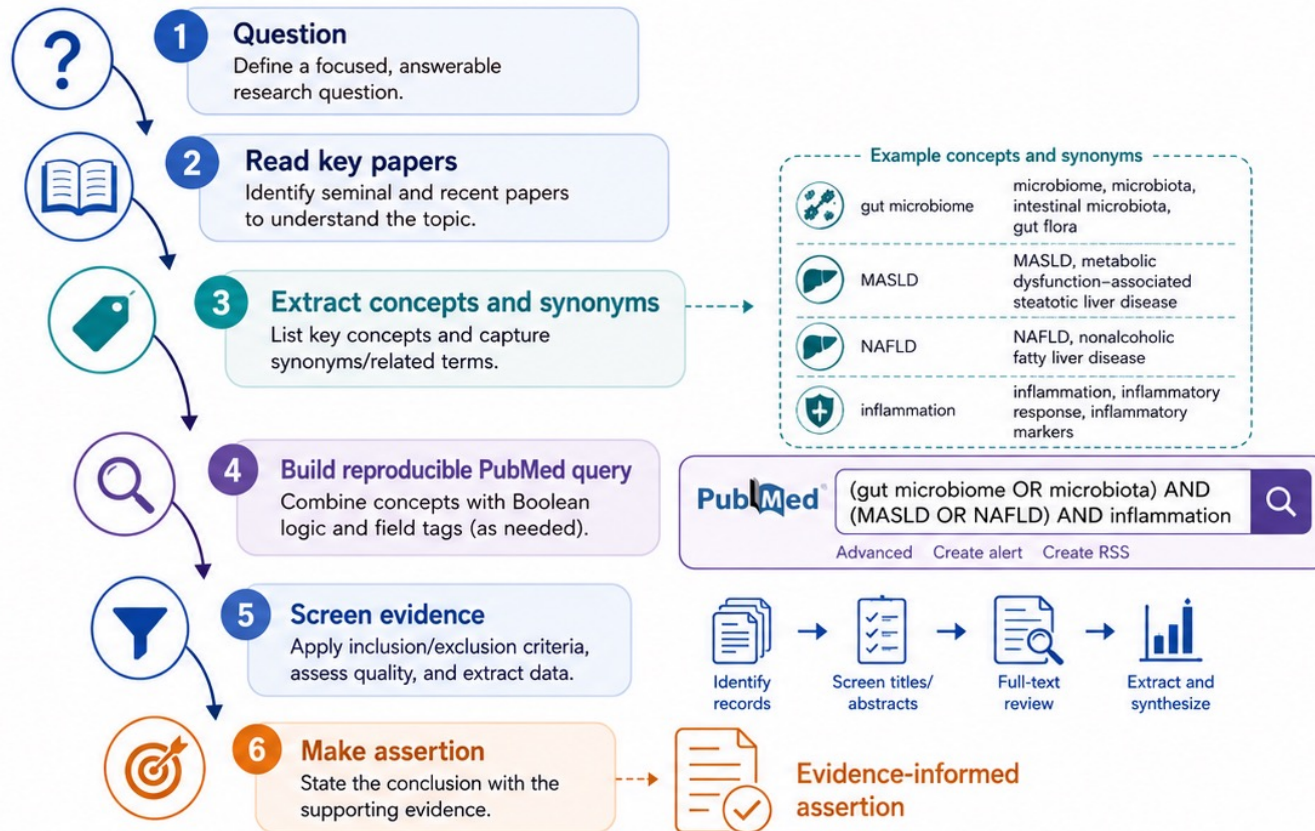
WHAT HAPPENS IF YOU SKIP THIS?



Core message

A precise question turns data exploration into hypothesis-driven science.

2. Why read papers and build reproducible PubMed queries before making assertions?



Benefits



Finds the real evidence base

Comprehensive and systematic searches capture the full landscape of relevant studies.



Reduces cherry-picking

Predefined queries and criteria limit selective use of studies that fit a narrative.



Lets others reproduce the search

Transparent queries and methods enable verification and build trust.



If skipped



Unsupported claims

Conclusions lack a solid evidence base.



Missed literature

Key studies may be overlooked.



Poor reproducibility

Others cannot replicate the search or findings.

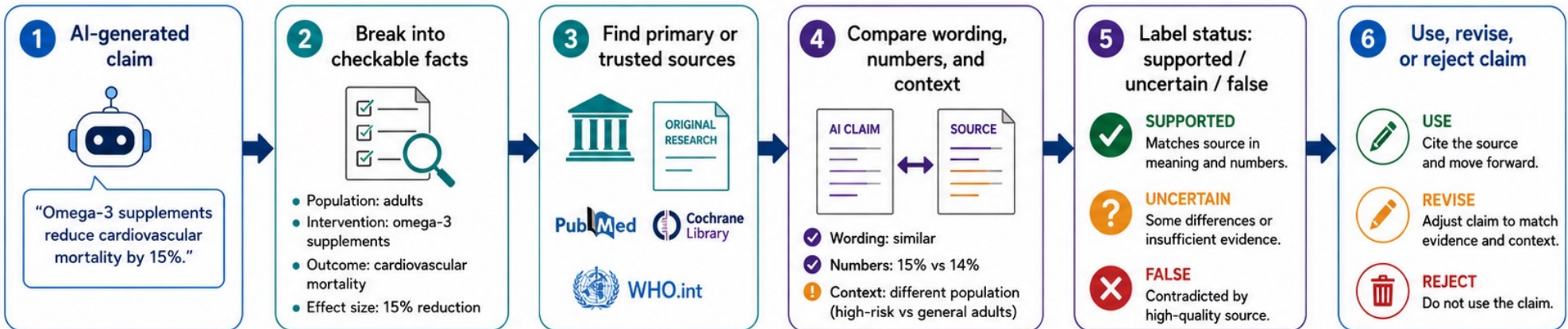


Core message

Assertions should be grounded in transparent, reproducible literature searches.



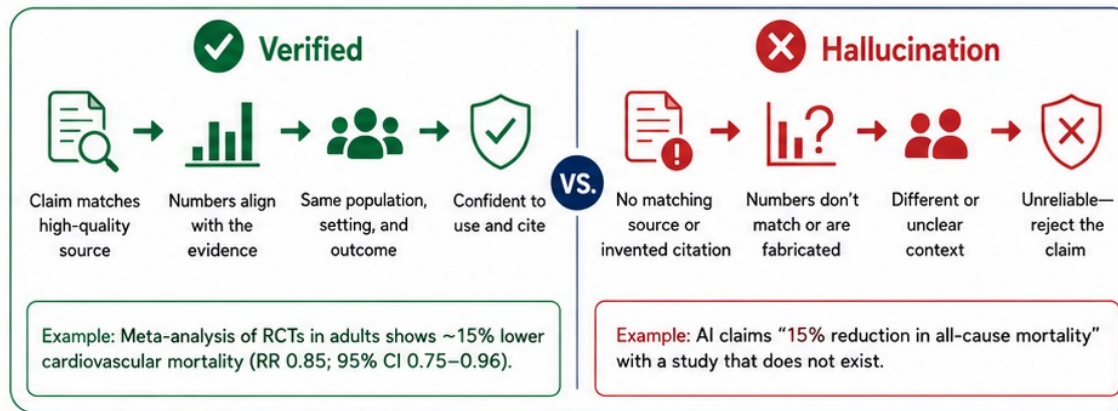
3. How to check every AI-generated claim against real sources



Check the date
Evidence changes. Always check when the source was published or last updated.

Prefer primary sources
Use original research, systematic reviews, and official reports over summaries or blogs.

Cite the source
Always record full citation (details, DOI/URL) so others can trace and verify it.



Quick verification checklist

- Who said it?
- Where is the source?
- Does the number match?
- Is the context the same?



Core message

AI can draft claims, but only source checking makes them reliable.

Every deliverable documents a step in scientific reasoning

Competency	What Students Should Be Able to Do	Concrete Deliverable
Question formulation	Convert a broad topic into a specific, testable, data-driven question	A six-level question funnel
Research design	Use PICO/PECO and feasibility criteria to define the study	A one-page analysis plan
Literature mining	Build PubMed queries with MeSH, Boolean operators, field tags	A documented search string with result count
AI-assisted synthesis	Use LLMs for expansion, screening, summarization, gap discovery	A prompt log and source-verification table
Reference management	Organize papers using Zotero or another reference manager	A library with 5 annotated papers

Two classes combine concept building with active workshops

Session 1

From Vague Interest to Testable Question

Main focus: Question formulation

Activity: Question funnel + PICO mini-workshop

Homework: Draft a data-driven question

Session 2

AI-Assisted Literature Mining

Main focus: Search, verify, organize

Activity: PubMed query clinic + AI prompt clinic

Homework: Find and annotate 3–5 papers

01

From Vague Interest to Testable Question

Session 1 | Scientific Question Formulation

Question formulation funnel | PICO/PECO/FINER frameworks

Minimal analysis plan | Common pitfalls | Workshop activity

Bioinformatics begins before code is written

The Scientific Pipeline



Without a precise question, large public datasets become traps: one can test many patterns, obtain impressive-looking figures, and still fail to produce a defensible scientific claim.

Core Principle

A good bioinformatics question is **specific, testable, biologically motivated, feasible** with available data, and **interpretable** through a planned workflow.

Vague questions produce weak science because they lack boundaries

Vague Question

"How does the microbiome affect health?"

Testable Question

"Are butyrate-producing taxa lower in T2D than controls, after adjusting for age and BMI?"

Risks of Unbounded Exploration

Data Dredging

Uncontrolled pattern searching after seeing the data

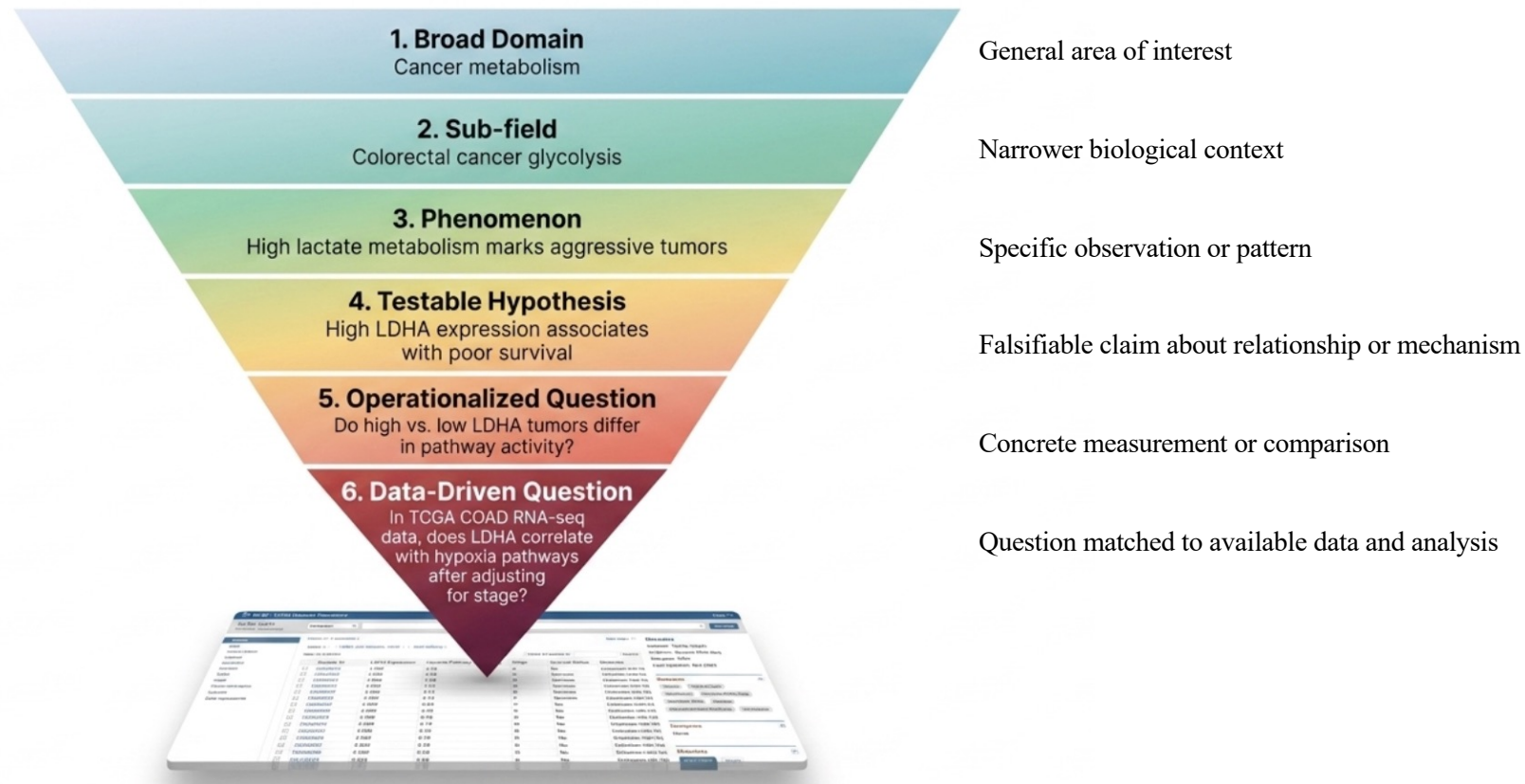
p-Hacking

Analytical flexibility producing false-positive significance

Multiple Testing

Thousands of features create false discovery risk

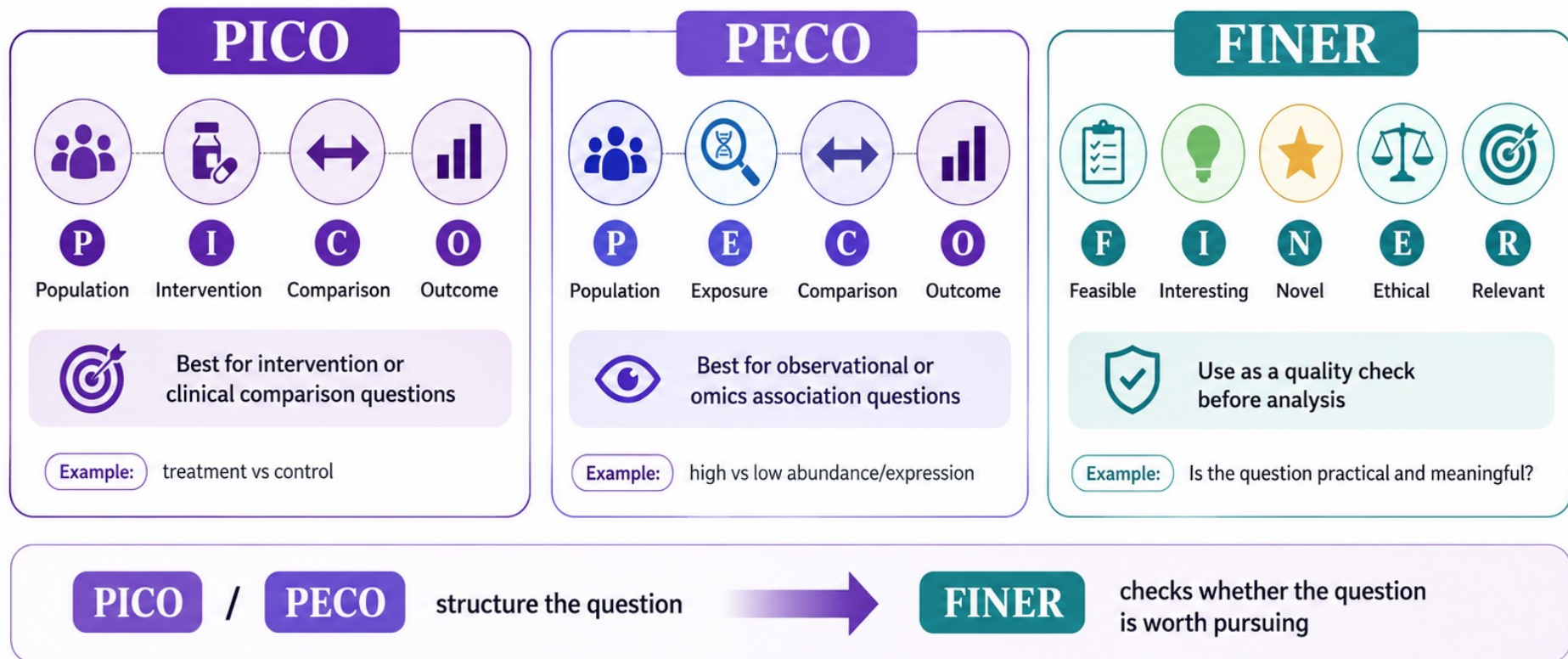
The question formulation funnel converts broad interest into a data-driven question



A stronger question names population, comparison, measurement, and data type

Broad Topic	Focused Issue	Data-Driven Question	Likely Analysis
Gut microbiota and obesity	Microbial metabolites and insulin resistance	Are butyrate pathways associated with fasting insulin after adjusting for diet and BMI?	Pathway profiling + regression
Cancer immunology	Single-cell tumor microenvironment	Are exhausted CD8+ T cell signatures enriched in high-myeloid tumors?	Cell annotation + module scoring
Antibiotic resistance	Resistance gene transfer	Are beta-lactam resistance genes enriched in hospital gut metagenomes?	AMR profiling + differential abundance
MAFLD/MASH biomarkers	Secreted proteins and severity	Which plasma proteins distinguish MAFLD from MASH after adjusting for BMI?	Differential abundance + ROC

PICO and PECO make biomedical questions structurally testable



Covariates determine whether an observed signal is biologically meaningful

Sample Metadata

Sample	Group	Age	Sex	BMI	Batch	Tissue	Depth
S01	T2D	58	M	31.2	B1	Stool	12M
S02	Control	45	F	24.8	B1	Stool	15M
S03	T2D	62	F	28.5	B2	Stool	11M

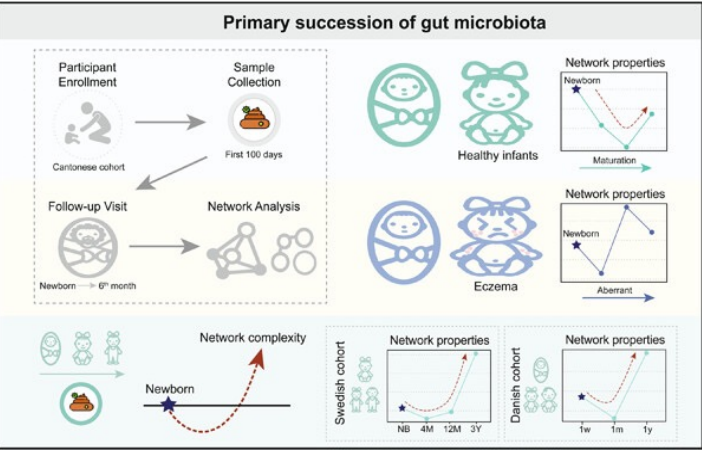
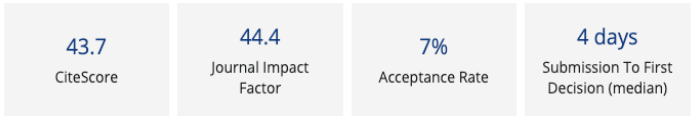
Key Confounders to Consider

Age	Sex	BMI	Batch	Seq Depth	Tissue Site
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Microbial network signatures of early colonizers in infants with eczema

Liujiang Huang^{1,2} | Guihua Pan² | Yifei Feng¹ | Zijiang Fan^{2,3} | Kai Ma⁴ | Runxin Wang⁴ | Guangxian Wang⁴ | Guangye Huang¹ | Sixia Huang¹ | Yuhui Hou¹ | Mulan Han² | Liwei Xie^{1,2,3} | Ying Ma¹

¹Obstetrics and Gynecology Medical Center, Zhujiang Hospital, Southern Medical University, Guangzhou, China
²Guangdong Provincial Key Laboratory of Microbial Culture Collection and Application, State Key Laboratory of Applied Microbiology Southern China, Institute of Microbiology, Guangdong Academy of Sciences, Guangzhou, China
³School of Public Health, Xixiang Medical University, Xixiang, China
⁴Jiangsu New-bio Biotechnology Co., Ltd., Jiangyin, China



	(ALL)	Control n = 22	Infantile_eczema n = 12	p Value
<i>Infants' characteristics</i>				
Gender				0.073
Female	17 (50.0%)	14 (63.6%)	3 (25.0%)	
Male	17 (50.0%)	8 (36.4%)	9 (75.0%)	
Birth weight (kg)	3.24 (0.31)	3.21 (0.31)	3.28 (0.30)	0.579
Height (cm)	50.4 (1.41)	50.2 (1.54)	50.6 (1.16)	0.455
Fetal position				0.372
LOA	30 (88.2%)	18 (81.8%)	12 (100%)	
LOT	1 (2.94%)	1 (4.55%)	0 (0.00%)	
ROA	3 (8.82%)	3 (13.6%)	0 (0.00%)	
Head girth (cm)	33.0 [32.0; 33.9]	33.0 [32.0; 33.4]	33.0 [32.0; 34.0]	0.569
Apgar 1 min				1
8	2 (5.88%)	1 (4.55%)	1 (8.33%)	
9	3 (8.82%)	2 (9.09%)	1 (8.33%)	
10	29 (85.3%)	19 (86.4%)	10 (83.3%)	
Apgar 5 min				0.353
9	1 (2.94%)	0 (0.00%)	1 (8.33%)	
10	33 (97.1%)	22 (100%)	11 (91.7%)	
Apgar 10 min: 10	34 (100%)	22 (100%)	12 (100%)	.
Feeding status				0.882
Exclusive breastfeeding	15 (44.1%)	9 (40.9%)	6 (50.0%)	
Predominant breastfeeding	19 (55.9%)	13 (59.1%)	6 (50.0%)	
<i>Maternal characteristics</i>				
Age (years)	30.0 [28.0; 33.0]	29.5 [28.0; 33.8]	30.0 [28.8; 31.5]	0.914
Prepregnancy BMI (kg m ⁻²)	20.7 (2.35)	21.1 (2.41)	19.8 (2.06)	0.104
Predelivery BMI (kg m ⁻²)	26.0 (2.95)	26.8 (2.82)	24.7 (2.77)	0.043
Weight gain (kg)	13.7 (4.97)	14.6 (4.63)	12.0 (5.34)	0.168
Fasting glycaemia (mmol L ⁻¹)	4.38 (0.29)	4.39 (0.33)	4.34 (0.21)	0.599
OGTT glucose1h (mmol L ⁻¹)	7.48 (1.61)	7.58 (1.55)	7.28 (1.78)	0.64
OGTT glucose2h (mmol L ⁻¹)	6.54 (1.12)	6.56 (1.16)	6.49 (1.11)	0.854
TC (mmol L ⁻¹)	6.16 (1.20)	6.34 (1.33)	5.84 (0.90)	0.226
TG (mmol L ⁻¹)	2.82 (1.09)	2.52 (0.98)	3.37 (1.11)	0.047

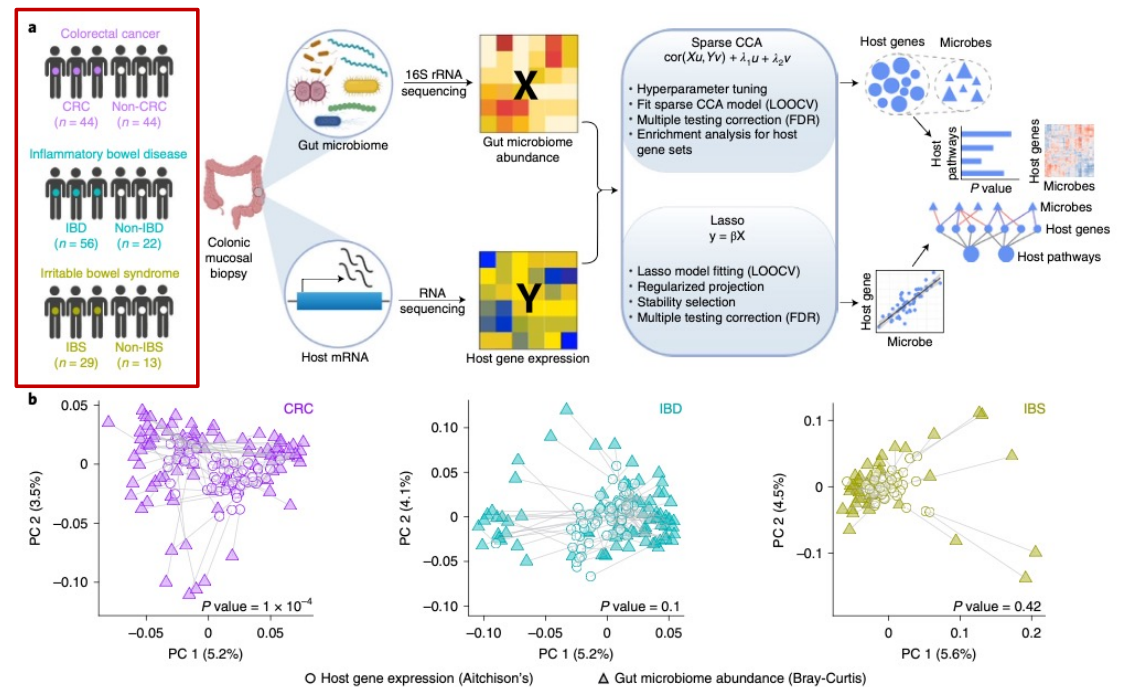


OPEN

Identification of shared and disease-specific host gene-microbiome associations across human diseases using multi-omic integration

Sambhawa Priya^{1,2}, Michael B. Burns³, Tonya Ward⁴, Ruben A. T. Mars⁵, Beth Adamowicz¹, Eric F. Lock⁶, Purna C. Kashyap⁵, Dan Knights^{4,7} and Ran Blehman^{1,8}✉

While gut microbiome and host gene regulation independently contribute to gastrointestinal disorders, it is unclear how the two may interact to influence host pathophysiology. Here we developed a machine learning-based framework to jointly analyse paired host transcriptomic ($n = 208$) and gut microbiome ($n = 208$) profiles from colonic mucosal samples of patients with colorectal cancer, inflammatory bowel disease and irritable bowel syndrome. We identified associations between gut microbes and host genes that depict shared as well as disease-specific patterns. We found that a common set of host genes and pathways implicated in gastrointestinal inflammation, gut barrier protection and energy metabolism are associated with disease-specific gut microbes. Additionally, we also found that mucosal gut microbes that have been implicated in all three diseases, such as *Streptococcus*, are associated with different host pathways in each disease, suggesting that similar microbes can affect host pathophysiology in a disease-specific manner through regulation of different host genes. Our framework can be applied to other diseases for the identification of host gene-microbiome associations that may influence disease outcomes.



A PRDM16-Driven Metabolic Signal from Adipocytes Regulates Precursor Cell Fate

Wenshan Wang,^{1,2} Jeff Ishibashi,^{1,2} Sophie Trefely,^{3,6} Mengle Shao,⁷ Alexis J. Cowan,⁸ Alexander Sakars,^{1,2} Hee-Woong Lim,^{1,4} Sean O'Connor,⁹ Mary T. Doan,⁹ Paul Cohen,⁹ Joseph A. Baur,^{1,2} M. Todd King,¹⁰ Richard L. Veech,¹⁰ Kyoung-Jae Won,^{1,4} Joshua D. Rabinowitz,¹ Nathaniel W. Snyder,¹ Rana K. Gupta,¹ and Patrick Seale^{1,2,11,*}

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¹¹Lead Contact

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<https://doi.org/10.1016/j.cmet.2019.05.005>

SUMMARY

The precursor cells for metabolically beneficial beige adipocytes can alternatively become fibrogenic and contribute to adipose fibrosis. We found that cold exposure or β 3-adrenergic agonist treatment of mice decreased the fibrogenic profile of precursor cells and stimulated beige adipocyte differentiation. This fibrogenic-to-adipogenic transition was impaired in aged animals, correlating with reduced adipocyte expression of the transcription factor PRDM16. Genetic loss of *Prdm16* mimicked the effect of aging in promoting fibrosis, whereas increasing PRDM16 in aged mice decreased fibrosis and restored beige adipose development. PRDM16-expressing adipose cells secreted the metabolite β -hydroxybutyrate (BHB), which blocked precursor fibrogenesis and facilitated beige adipogenesis. BHB catabolism in precursor cells, mediated by BDH1, was required for beige fat differentiation *in vivo*. Finally, dietary BHB supplementation in aged animals reduced adipose fibrosis and promoted beige fat formation. Together, our results

demonstrate that adipocytes secrete a metabolite signal that controls beige fat remodeling.

INTRODUCTION

Brown and beige adipocytes burn energy for heat production and have garnered much attention because of their capacity to counteract metabolic disease (Harms and Seale, 2013; Kajimura et al., 2015). Brown adipocytes are located in discrete deposits of brown adipose tissue (BAT), whereas beige adipocytes develop in white adipose tissue (WAT) in response to environmental cold and other stimuli (Wang and Seale, 2016). Brown and beige adipocytes have abundant mitochondria that contain uncoupling protein 1 (UCP1) (Cannon and Nedergaard, 2004). Upon activation by cold or β -adrenergic signaling, UCP1 catalyzes proton leak across the inner mitochondrial membrane, resulting in increased oxygen consumption and heat production (Cannon and Nedergaard, 2004; Fedorenko et al., 2012; Klingenberg and Huang, 1999; Ricquier, 2011).

Increasing the amount and/or activity of beige adipocytes in mice promotes a lean and healthy metabolic phenotype (Wu et al., 2013). Conversely, reductions in beige fat activity predispose animals to the harmful metabolic effects of a high-fat diet (Cohen et al., 2014). Beige adipocytes develop from

PDGFR α + precursor cells that are also capable of differentiating into fibrosis-inducing myofibroblast cells (Berry et al., 2016; Lee et al., 2012; Lin et al., 2018; Long et al., 2014; Marcelin et al., 2017; McDonald et al., 2015; Shao et al., 2016; Wang et al., 2013). Analogous mesenchymal cell populations possessing adipogenic and fibrogenic potential, called fibro-adipogenic progenitor (FAP) cells, are also present in skeletal muscle and heart (Joe et al., 2010; Lemos et al., 2015; Uezumi et al., 2010).

Myofibroblast differentiation in adipose tissue is stimulated by various pathways, including hypoxia/HIF1 α and TGF β (Halberg et al., 2009; Marcelin et al., 2017; Sun et al., 2013a, 2013b; Vila et al., 2014). HIF1 α activation in adipose tissue is sufficient to trigger fibrosis, whereas pharmacological inhibition of HIF1 α reduces fibrosis and improves systemic metabolism (Halberg et al., 2009; Jiang et al., 2011; Sun et al., 2013a). TGF β promotes fibrogenesis through activation of Rho-associated protein kinase (ROCK) and resultant assembly of actin stress fibers (Carthy, 2018). Notably, deletion of *Mrtfa*, a downstream effector in the TGF β pathway, blocks the myofibrogenic programming of adipose precursors and facilitates beige fat differentiation (Lin et al., 2018; McDonald et al., 2015). However, the upstream signals that regulate the fibrogenic versus adipogenic activity of precursor cells during beige fat development are yet unknown.

The zinc-finger transcriptional factor PRD1-BF1-RIZ1 homologous domain-containing protein 16 (PRDM16) is a powerful driver of brown and beige adipocyte identity (Cohen et al., 2014; Harms et al., 2015; Ishibashi and Seale, 2015; Kajimura et al., 2008; Ohno et al., 2013; Seale et al., 2007, 2011). PRDM16 co-activates PPAR γ and PPAR α in adipocytes to activate the expression of thermogenic genes (Hondares et al., 2011; Seale et al., 2008). Interestingly, PRDM16 overexpression in adipose tissue reduces high-fat-diet-induced fibrosis (Hasegawa et al., 2018). However, if and how PRDM16 action in adipocytes affects the fibrogenic versus beige adipogenic fate of precursor cells was unclear.

In this study, we identify a PRDM16-regulated adipocyte-to-precursor paracrine signal that suppresses fibrogenesis and facilitates beige fat development. Specifically, PRDM16 drives a fatty acid oxidation and ketogenesis program in adipose cells, leading to secretion of the metabolite β -hydroxybutyrate (BHB). BHB acts on precursor cells to block HIF1 α - or TGF β -induced myofibroblast differentiation and promote beige adipocyte differentiation. This action of BHB in precursor cells is dependent on the ketolytic enzyme BDH1, revealing an important role for ketone metabolism in adipose tissue remodeling. Finally, raising PRDM16 or BHB levels in mice reverses aging-induced fibrosis and restores beige fat developmental potential.

RESULTS

Aging Impairs Beige Adipogenesis and Promotes Fibrogenesis

Human brown fat activity levels decline with aging, correlating with a decrease in metabolic rate and increase in fat mass (Cypess et al., 2009; Ouellet et al., 2011; Pfannenberger et al., 2010; Yoneshiro et al., 2011). The development of UCP1+ beige adipocytes in response to cold or the β 3 agonist CL316,243 (CL) was drastically reduced in aged (12-month-old) compared to young (2-month-old) mice (Figures 1A, 1B, and S1A–S1D), consistent with the results from other studies (Berry et al., 2017; Rogers et al., 2012). Beige fat cells in iWAT develop via the activation of thermogenic genes in mature adipocytes (i.e., beige conversion of mature adipocytes) or through the beige adipogenic differentiation of precursor cells (Berry et al., 2016; Lee et al., 2012, 2015; Shao et al., 2016; Wang et al., 2013). To determine if aging reduces the differentiation of beige adipocytes, we compared the beige response in young and aged *AdipoChaser* mice (*Adipoq^{Cre}*; *TRE^{Cre}*; *Rosa26^{YFP}*) (Wang et al., 2013). Doxycycline (dox) treatment of these mice induced GFP expression in mature (PLIN1+) adipocytes (pulse) (Figures S1E and S1F). After CL treatment (chase), the iWAT of young mice contained large areas of GFP(–) (i.e., new) multilocular beige adipocytes, with the proportion of GFP(–) adipocytes increasing from ~5% to >35% (Figures S1F and S1G). By contrast, CL treatment did not increase the proportion of unlabeled adipocytes in old mice, indicative of impaired adipogenesis (Figures S1F and S1G).

We used RNA-seq to profile iWAT gene expression in young and aged mice, treated with either vehicle or CL. iWAT from aged mice displayed a blunted upregulation of many thermogenic genes in response to CL treatment, including genes involved in oxidative phosphorylation and electron transport chain activity (Figure S1H). CL treatment also downregulated the expression of extracellular matrix (ECM) and fibrosis marker genes in young iWAT, whereas these genes were unchanged or increased in iWAT from aged animals (Figures 1C–1E and S1I). These fibrosis-signature genes (e.g., *Col1a1*, *Col3a1*, *Lox1*, *Fbn2*) were expressed at higher levels in stromal-vascular cells (SVCs; a population enriched for precursor cells) than in mature adipocytes, and their expression levels in SVCs were decreased by cold exposure (Figures 1F and S1J). Finally, CL treatment decreased the expression of fibrosis genes in SVCs from young but not aged mice (Figure 1G). At a histological level, CL treatment decreased collagen deposition and fibrosis in the iWAT of young mice but augmented fibrosis in aged animals (Figure 1H). Notably, aging did not impair BAT activation or induce

A minimal analysis plan prevents uncontrolled exploration

Dataset

Which cohort or database?

Inclusion

Which samples in or out?

Comparison

What exactly is compared?

Covariates

What confounders to adjust?

Method

DESeq2, Cox regression,
PERMANOVA, Wilcoxon

Visualization

PCA, volcano plot, survival curve,
heatmap

Validation

Independent cohort, sensitivity analysis

Key question: Can another student inspect your plan and detect weak design *before* you start analyzing? If yes, the plan is working.

Most weak bioinformatics projects fail at the question stage

Pitfall	Why Problematic	Weak Version	Better Version
Too broad	No single dataset can answer it	<i>"How does the microbiome affect health?"</i>	"Which microbial pathways differ between T2D and controls?"
Not testable	No measurable variable	<i>"Is inflammation important?"</i>	"Are IL6/TNF scores higher in diseased tissue?"
Confounded	Signal reflects other variables	<i>"Are obese and lean microbiomes different?"</i>	"Are differences present after adjusting for age, sex, diet?"
Reverse causality	Direction is unclear	<i>"Does cancer cause gene X expression?"</i>	"Is high gene X expression associated with stage?"
Trivial	No meaningful novelty	<i>"Does TP53 mutate in cancer?"</i>	"Does TP53-mutant CRC show a distinct immune signature?"
Not feasible	Required data do not exist	<i>"Analyze proteome of every neuron in 10,000 brains"</i>	"Use available snRNA-seq to profile selected cell types"

Students can improve any topic by moving it through six funnel levels

Think–Pair–Share

Choose One Topic

- Gut microbiota and metabolic disease
- Single-cell transcriptomics in cancer immunology
- Antibiotic resistance genomics
- Epigenetic regulation in neurological disorders
- Muscle regeneration and aging
- Proteomic biomarkers in MAFLD/MASH

Worksheet Steps

#	Student Action	Output
1	Write a broad topic	One phrase
2	Narrow to a phenomenon	One sentence
3	State a testable hypothesis	One falsifiable claim
4	Apply PICO/PECO	Four structured elements
5	Identify a dataset or data type	One database or accession
6	Draft a minimal analysis plan	Data, method, output, validation

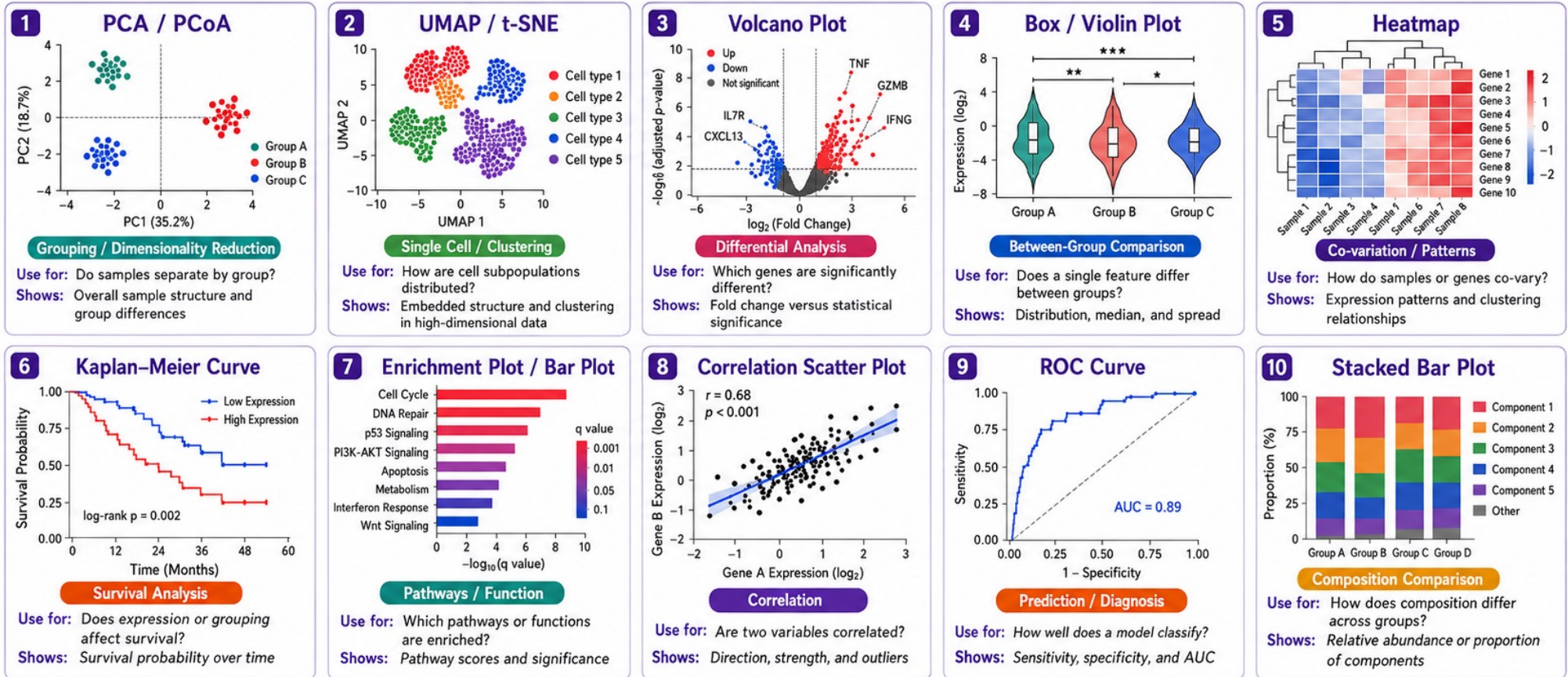
A good question already implies the figure that should answer it

Question Type	Expected Figure	What It Shows
"Do groups cluster differently?"	PCA / PCoA plot	Sample separation by group
"Which genes are differentially expressed?"	Volcano plot	Fold change vs significance
"Does expression predict survival?"	Kaplan-Meier curve	Survival probability over time
"What pathways are enriched?"	Enrichment / bar plot	Pathway scores and significance
"How do samples co-vary?"	Heatmap	Gene expression across samples

Class exercise: Name the figure you would expect from your Workshop question.

Common Research Questions, Plot Types, and When to Use Them

— A good question already implies the figure that should answer it —



Core principle: Define the question first, then choose the plot that best answers it.



02

AI-Assisted Literature Mining and Verification

Session 2 | Search, Verify, Organize

PubMed search strategy | MeSH and Boolean logic | AI-assisted mining

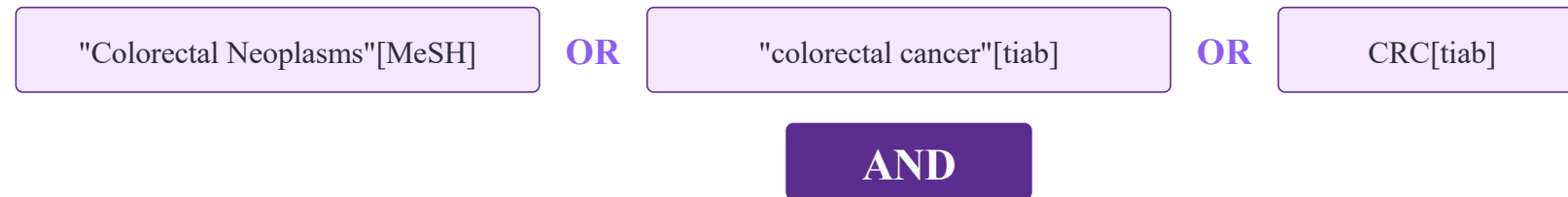
Citation hallucination | Source verification | Zotero workflow

Different literature tools solve different parts of the evidence workflow

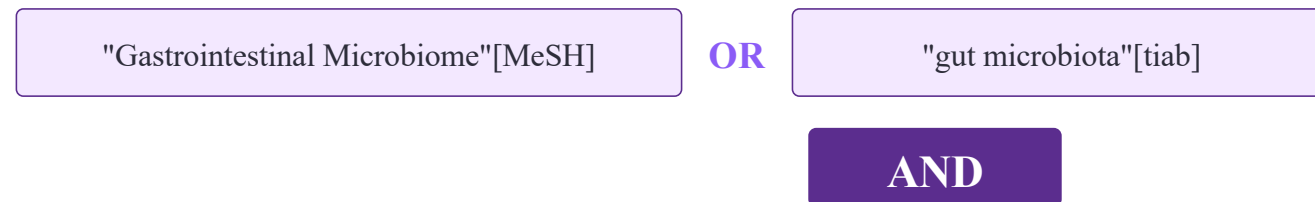
Tool	Strength	Best Use	Caution
PubMed	Biomedical specificity, MeSH terms, advanced syntax	Primary biomedical search	Does not cover all disciplines
Google Scholar	Broad coverage and citation counts	Finding highly cited papers	Ranking is less transparent
Semantic Scholar	AI-powered search, TLDR, citation context	Related papers + quick summaries	AI summaries must be checked
Elicit	AI-native extraction and literature tables	Systematic question-driven screening	Not a full systematic review
Perplexity	Web-connected synthesis with links	Fast orientation on new topics	Source quality varies; verify
Zotero	Reference storage, PDF annotation, citation	Course reference library	Requires consistent habits

A PubMed query should be built concept by concept

Concept 1: Disease



Concept 2: Microbiome



Concept 3: Method



Limits

Date:

"2020"[dp]:"2026"[dp]

Type: review[pt] or clinical
trial[pt]

Species: humans[MeSH]

MeSH, Boolean operators, and field tags make search reproducible

Syntax	Meaning	Example	Teaching Note
AND	Both concepts must appear	cancer AND microbiome	Narrows results
OR	Either synonym may appear	microbiome OR microbiota	Broadens within a concept
NOT	Exclude a concept	cancer NOT review	Use sparingly; may remove relevant papers
[tiab]	Search title and abstract	"gut microbiota"[tiab]	Useful for new terms not well indexed
[MeSH]	Search Medical Subject Heading	"Gastrointestinal Microbiome"[MeSH]	Controlled vocabulary improves precision
[dp]	Date of publication	"2020"[dp]:"2026"[dp]	Filter by publication date window
[pt]	Publication type	review[pt] or clinical trial[pt]	Filter by evidence type

Record every search string and result count so the workflow can be reproduced

Search strategy is iterative rather than final on the first attempt

Worked Example: CRC + Microbiome + RNA-seq

```
("Colorectal Neoplasms"[MeSH] OR "colorectal cancer"[tiab] OR CRC[tiab]) AND ("Gastrointestinal Microbiome"[MeSH] OR "gut microbiota"[tiab] OR "gut microbiome"[tiab]) AND ("RNA-Seq"[tiab] OR transcriptomics[tiab]) AND ("2020"[dp]:"2026"[dp])
```

Refinement Decision Guide

Too Many Results?

Add a tissue, method, species, or publication-type filter.
Consider adding a concept with AND to narrow further.

Too Few Results?

Remove one concept or broaden synonyms within each OR group. Check whether terms are well indexed in MeSH.

AI can accelerate literature mining but cannot replace judgment



Human Checkpoints at Every Step

AI Task	Good Prompt Pattern	Human Verification
Query expansion	Generate synonyms, MeSH terms, variants	Check terms in MeSH browser; run test searches
Abstract screening	Classify as include/exclude/uncertain	Manually review uncertain and high-impact papers
Paper summarization	Question, design, methods, findings, limitations	Read abstract; check summary matches the paper
Gap discovery	Identify consensus, controversy, gaps	Confirm gaps with recent reviews and primary studies
Hypothesis generation	Convert gap to PICO/PECO and analysis plan	Assess feasibility and novelty before claiming

Citation hallucination is the most dangerous AI error in academic work

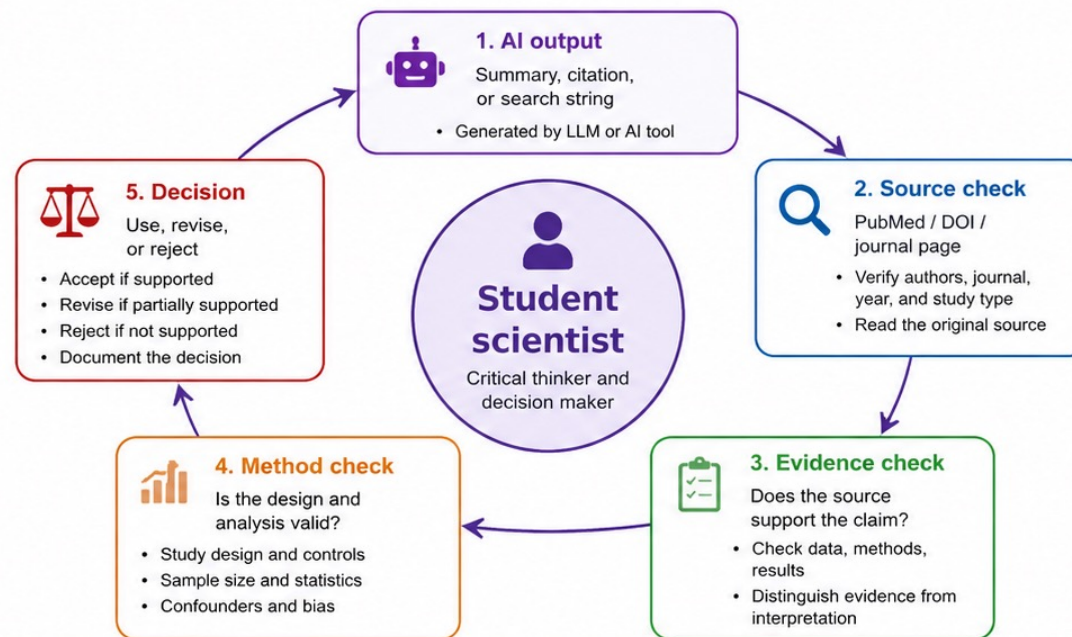
Risk	Example	How to Detect	Required Action
Fake citation	A plausible DOI that does not resolve	Search PubMed, Crossref, journal site	Remove or replace with verified source
Misattributed claim	AI says causality; study was observational	Read abstract, methods, key figures	Downgrade claim to association
Wrong method	AI says DESeq2; paper used limma	Check methods section	Correct summary; note exact method
Outdated consensus	AI repeats old view despite new studies	Search recent reviews and 2024–2026 papers	Update synthesis; note controversy
Overconfident gap	AI claims nobody has studied a topic	Search exact gap terms and citation networks	Use cautious wording: "limited evidence"

Rule: No citation enters your work unless you verified it exists and read at least the abstract.

Verification intensity should match scientific risk

Claim Type	Risk	Verification Method	When to Do It
General explanation	Low	Check against textbook or lecture notes	Quick mental check
R syntax or function name	Low	Run help() or check documentation	Before running code
Software parameters	Medium	Read official vignette or manual	Before writing methods section
Statistical design choice	Medium	Compare with published best practices	Before finalizing analysis plan
Biological mechanism claim	High	Read primary source + independent review	Before including in any report
Citation, DOI, or data accession	High	Resolve DOI, search PubMed, open journal page	Always, before citing

Human-in-the-loop verification protects scientific reliability

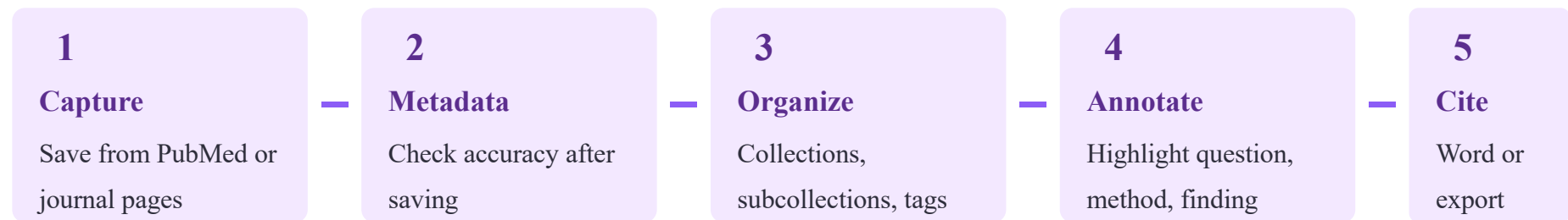


When in doubt, do not cite. Verify, then decide.

Source quality determines how strongly a paper should influence conclusions

Criterion	Stronger Evidence	Weaker Evidence
Study design	Clear cohort, controls, covariates, validation	Vague sample description or no controls
Sample size	Reasonable power for the analysis type	Very small n without justification
Data availability	Data in GEO/SRA/ENA/TCGA with accession	No accession number or unclear availability
Code transparency	Code, versions, parameters documented	Only high-level method names provided
Statistics	Multiple-testing correction, effect sizes	Only unadjusted p-values or unclear tests
Reproducibility	Independent cohort or sensitivity analysis	Single dataset with no validation

Zotero turns literature search into a reproducible reference library



Recommended Habits

Action	Recommended Habit
Create a course collection	Name it "Bioinformatics Week 2" for easy access
Create subcollections	Separate: question formulation, PubMed, AI mining, topic papers
Add tags	Use structured tags: method: RNA-seq; disease: CRC; evidence: review
Annotate PDFs	Highlight research question, dataset, method, finding, limitation

LLMs answer questions, but agents perform controlled scientific workflows

LLM Chatbot

Answers questions, generates text, drafts ideas. No tools, no memory across sessions, no file access.

AI Agent

Uses tools, accesses files, follows skills, maintains logs. Converts instructions into repeatable workflows.

Agent Components

Model + Instructions

Tools + Permissions

Memory + Logs

Safety rule: An agent can accelerate search and analysis, but a human must approve claims, methods, and citations.

Prompt clinics turn vague AI use into reproducible support

Before

"Find papers about microbiome and diabetes."

After

"Act as a literature-mining assistant. Task: build a PubMed strategy on gut microbiome and T2D. Requirements: use MeSH, Boolean operators, [tiab], [dp]; separate concepts; explain each block. Then find 3–5 recent papers (2020–2026) and summarize each."

Why the Second Prompt Is Better

Role + Task

Names the AI role and the exact task to perform

Requirements

Specifies tools, syntax, and constraints explicitly

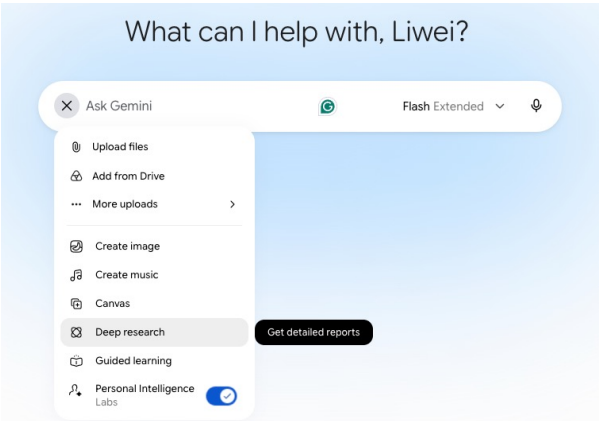
Verification

Asks AI to list claims needing source verification

AI tools for literature search

Lastest research summary in depth of Thbs4 in health and diseases, specifically in aging, metabolic diseases and the pro and con of Thbs4 in health and diseases

https://gemini.google.com/app



Thrombospondin-4 (THBS4) is a pentameric matricellular glycoprotein of the thrombospondin family that functions as a critical mediator of cell-to-cell and cell-to-matrix interactions (Genaro & Luo, 2024). Under normal physiological conditions, baseline THBS4 expression is low in adult somatic tissues but surges dramatically in response to tissue injury, biomechanical stress, inflammation, and cellular remodeling (Stenina-Adognravi & Plow, 2019).

Role in Aging and Tissue Integrity

- Synaptogenesis and Cognitive Rejuvenation:** THBS4 acts as a key circulating factor enriched in young systemic circulation ("young blood") that directly targets neurons to induce dendritic arborization, spine formation, and synaptic NMDA receptor recruitment, counteracting age-related cognitive decline (Gan & Südhof, 2019). In the aging human brain, *THBS4* variants maintained by balancing selection interact with sex to modulate gray matter volume and neuroinflammatory responses in Alzheimer's disease (Cagliani et al., 2013).
- Musculoskeletal and Cardiac Aging:** THBS4 stabilizes the sarcolemma by organizing membrane attachment complexes. Deletion of *Thbs4* causes spontaneous dystrophic muscle degeneration with aging and exacerbates cardiomyopathy, whereas overexpressing THBS4 protects skeletal muscle and heart membranes against strain-induced damage (Vanhoutte et al., 2016).

Role in Metabolic and Cardiovascular Diseases

- Adipose Tissue and Vascular Inflammation:** In metabolic disease models, THBS4 drives macrophage recruitment into visceral and perivascular adipose tissue. Deletion of *Thbs4* reduces adipose tissue chemokine production, suppresses macrophage accumulation in metabolic fat depots and arterial walls, and prevents diet-induced neointimal hyperplasia and vascular stenosis (Lv et al., 2016).
- Endoplasmic Reticulum (ER) Stress:** Intracellularly, THBS4 triggers an adaptive unfolded protein response (UPR) via ATF6 α activation, inducing ER expansion and protecting cardiomyocytes and smooth muscle cells from metabolic or biomechanical stress (Genaro & Luo, 2024).

Here's a research plan for that topic. If you need to update it, let me know!

THBS4 in Health and Disease

Research Websites

(1) Search for current scientific literature and review articles on Thrombospondin-4 molecular structure, baseline expression, and general physiological function in health and tissue homeostasis.

(2) Investigate recent findings regarding THBS4 in aging, specifically its effects on central nervous system synaptogenesis, cognitive rejuvenation, muscle sarcolemma stability, and tissue integrity.

(3) Search for research on THBS4 in metabolic disorders, focusing on its mechanisms in adipose tissue macrophage recruitment, metabolic vascular inflammation, and endoplasmic reticulum stress pathways.

(4) Research other major pathological contexts involving THBS4, including tissue fibrosis, cardiovascular disease, tumor microenvironment regulation in cancer, and neuropathic pain.

(5) Evaluate and compare the dual roles (pros and cons) of THBS4, highlighting where its upregulation is protective versus where it drives pathological remodeling or disease progression.

(6) Synthesize the gathered data into an in-depth summary covering THBS4 biology, aging impact, metabolic roles, and overall health outcomes.

Analyze Results

Create Report

Ready in a few mins

Edit plan

Start research

THBS4 in Health and Disease

Show thinking

Mapping Core Physiological Functions

I am initiating a comprehensive investigation into Thrombospondin-4 (THBS4), focusing on its molecular structure, baseline expression patterns, and fundamental physiological roles in maintaining tissue homeostasis. To establish a solid foundation, I am synthesizing current literature on its domain architecture and receptor interactions, particularly how it engages with key cell-surface proteins to facilitate intercellular signaling and extracellular matrix assembly.

Evaluating Role in Aging and Neurobiology

I am examining the specific involvement of THBS4 in aging and tissue integrity, with a particular focus on the central nervous system and skeletal muscle. My goal is to resolve how THBS4 influences synaptic formation, potential cognitive rejuvenation, and sarcolemmal stability in muscle tissue, while mapping out its distinct mechanisms of action across different physiological contexts.

Investigating Disease Pathways and Next Steps

To address major information gaps surrounding THBS4's dual nature, I am structuring searches across its involvement in metabolic dysregulation, ER stress pathways like ATF6-alpha, cardiovascular remodeling, fibrosis, neuropathic pain, and tumor microenvironments. My next immediate step is to execute targeted literature searches to gather data on these diverse pathological pathways and analyze the context-dependent shift between its protective and maladaptive effects.

pmc... Pathophysio...

pmc... Thrombospon...

pmc... Control of Org...

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pub... Thrombospon...

ncbi... 7060 - Gene R...

pmc... Thrombospon...

ncbi... 21828 - Gene R...

Dual biological roles

Analyzing recent literature reveals that thrombospondin-4 operates through both extracellular and intracellular pathways. Beyond its traditional extracellular matrix interactions with integrins and receptors like CACNA2D1 during synaptogenesis and pain signaling, it acts internally within the endoplasmic reticulum. Secreted and non-secreted variants engage the adaptive unfolded protein response via ATF6 activation, which expands cellular vesicles and confers protection against cardiac stress and muscular dystrophy.

Pathological contrast and disease mechanisms

Resolving its distinct physiological roles highlights a clear contrast with other family members. Thrombospondin-4 uniquely promotes angiogenesis and tissue remodeling rather than inhibiting it. In pathological contexts, high expression in cancer-associated fibroblasts correlates with tumor progression across multiple malignancies, while its involvement in adipose tissue inflammation connects it directly to metabolic stress and vascular remodeling.

Next research steps

To build a complete mechanistic map, I am planning targeted inquiries into its precise receptor interactions during central nervous system synaptogenesis, its specific role in metabolic macrophage recruitment, and the exact signaling cascades driving its pro-tumoral effect in the extracellular microenvironment.

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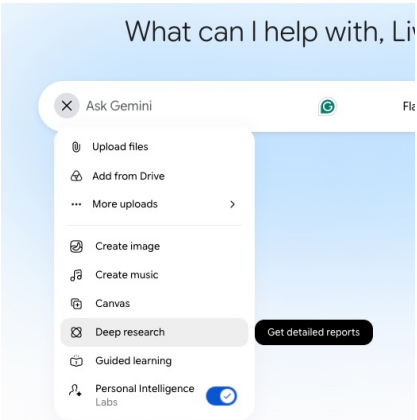
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AI tools for literature search

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Structural Dynamics, Molecular Mechanisms, and Pathophysiological Roles of Thrombospondin-4 in Aging, Metabolic Disease, and Tissue Homeostasis

Thrombospondin-4 (THBS4) is a soluble, multimeric matricellular glycoprotein belonging to Subgroup B of the thrombospondin protein family. Unlike classical structural extracellular matrix (ECM) scaffold components such as fibrillar collagens or laminins, THBS4 functions primarily as a dynamic mediator of cell-to-cell and cell-to-matrix communication, cellular migration, proliferation, and organellar adaptation. Expressed abundantly during embryonic organogenesis, THBS4 is largely silenced in healthy adult tissues, yet exhibits remarkable transcriptional plasticity by undergoing rapid re-expression in response to mechanical shear stress, tissue trauma, metabolic overload, and pathological remodeling.

Molecular and genetic studies have delineated the versatile roles of THBS4 across diverse physiological domains, establishing it as an essential regulator of central nervous system synaptogenesis, muscle sarcolemmal integrity, microvascular lipid processing, and systemic tissue repair. However, the biological activity of THBS4 is fundamentally double-edged. While its intracellular organellar signaling and localized matrix deposition promote adaptive repair and structural stabilization, its chronic or dysregulated accumulation drives pathological vascular inflammation, central pain sensitization, adipose tissue insulin resistance, and tumor-promoting stroma activation.

Structural Architecture, Biosynthesis, and Signal Transduction Networks of Thrombospondin-4

Molecular Domain Architecture and Genetic Polymorphisms

Human THBS4 is encoded by the *THBS4* gene located on chromosome 5q14.1. Synthesized as a 140 kDa precursor polypeptide, mature THBS4 assembles into a 700 kDa homopentameric complex. This pentameric organization distinguishes Subgroup B thrombospondins (THBS3, THBS4, and THBS5/COMP) from Subgroup A members (THBS1 and THBS2), which assemble exclusively as trimers. The modular domain organization of each THBS4 subunit extends sequentially from its amino-terminus to its carboxy-terminus.

At the N-terminal region, a dedicated domain is tethered to an oligomerizing coiled-coil motif where interchain disulfide bonds lock five identical polypeptide chains into a functional, star-shaped pentameric structure. Positioned downstream of the coiled-coil domain are four Type 2 epidermal growth factor-like (EGF-like) repeats, which harbor critical interaction sites for cell-surface receptors and house clinically relevant genetic variations. Following the EGF-like repeats lies a sequence of thirteen Type 3 calcium-binding repeats, structurally designated as the "calcium-binding wire module". Cooperative binding of calcium ions (Ca^{2+}) to these Type 3 repeats induces a major conformational shift that rigidifies the tertiary fold, stabilizing the molecule and exposing functional binding epitopes. The protein terminates in a carboxy-terminal globular domain structurally homologous to L-type lectins. Together with the EGF-like and Type 3 repeats, this C-terminal region forms the highly conserved "signature domain" responsible for tethering THBS4 to ECM macromolecules and integrin receptors.

A nonsynonymous single nucleotide polymorphism (SNP) in human *THBS4*, resulting in an alanine-to-proline substitution at position 387 (A387P; rs1866389 or 1186C), resides within the third EGF-like repeat. The Pro387 variant alters local secondary structure and accelerates Ca^{2+} dissociation kinetics. Epidemiological and functional investigations demonstrate that homozygosity for the P387 variant represents an independent risk factor for premature myocardial infarction, coronary artery disease, and accelerated atherogenesis, primarily due to altered structural flexibility, enhanced macrophage recruitment, and heightened endothelial inflammatory responses.

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AI tools for literature search

https://chatgpt.com/

20206 Fall Bioinformatics cla...

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🔭 **Deep research** Lastest research especially with in 3 to 5 years and most recent research. Provide complete summary and in depth analysis of Thbs4 in health and diseases, specifically in aging, metabolic diseases and the pro and con of Thbs4 in health and diseases

PubMed Buddy
Worked for 2m 1s >

THBS4/TSP4 in health, aging, and disease: an updated synthesis through August 2026

I reviewed the recent THBS4 literature with emphasis on **2023–2026 primary studies**, supplemented by older mechanistic papers where they remain essential for interpreting the newer findings. The central conclusion is that **THBS4 is not intrinsically "protective" or "pathogenic."** It is a **stress-responsive matricellular protein whose effect depends strongly on cell type, tissue, subcellular location, timing, and disease stage**. This duality is especially important when thinking about aging and metabolic disease.

THBS4 encodes thrombospondin-4, a pentameric, calcium-binding member of the thrombospondin family. It participates in cell–matrix communication, ECM organization, migration, angiogenesis, synaptogenesis, and tissue remodeling. Unlike a conventional structural ECM protein, THBS4 can also function intracellularly, particularly in the endoplasmic reticulum, where it participates in adaptive stress responses. Human expression is relatively enriched in adipose tissue and heart, although cellular sources differ greatly by tissue and disease state.

The most useful conceptual model

A good way to understand THBS4 is:

acute stress / repair → THBS4 often adaptive and protective

but

persistent stromal activation / fibroproliferation → THBS4 can become maladaptive and disease-promoting.

That simple model explains many apparently contradictory publications.

For example, THBS4 can enhance adaptive ER-stress signaling and membrane stability in cardiomyocytes, promote bone repair angiogenesis, support wound healing, preserve the aortic wall in aneurysm, and enhance cytotoxic T-cell function. Yet in other settings it promotes pulmonary vascular remodeling, renal fibrosis, pulmonary fibrosis, neuropathic synaptogenesis, and therapy resistance in cancer.

12. Most important recent papers to read first

Year	Study	Why it matters	DOI	📄
2026	Coenen et al., <i>Aging Cell</i>	ECM aging clock; THBS4 associated with decelerated ECM aging	10.1111/ace.170474 (+) IF: 7.7 Q1	
2026	Zhang et al., <i>World J Gastroenterol</i>	Exercise/MASLD: muscle and circulating THBS4 decrease after exercise	10.3748/wjg.v32.i13.11398 5 (+)	
2026	Zeng et al., <i>Hypertension</i>	THBS4–TGF-β/SMAD2 positive-feedback loop in pulmonary hypertension	10.1161/HYPERTENSIONA.125.25968 (+) IF: 9.8 Q1	
2026	Blanco-Casoliva et al., <i>Br J Pharmacol</i>	THBS4 unexpectedly protective in abdominal aortic aneurysm	10.1111/bph.70402 (+) IF: 7.5 Q1	
2026	Zhou et al., <i>Int Immunopharmacol</i>	THBS4 implicated in pulmonary fibrosis	10.1016/j.intimp.2025.115782 (+) IF: 5.6 Q1	
2026	Shu et al., <i>Immunity Inflamm Dis</i>	THBS4 protective in disc cells via PI3K/AKT	10.1002/jid3.70484 (+) IF: 3.5 Q2	
2025	Song et al., <i>Bone Research</i>	THBS4 drives angiogenesis in bone repair	10.1038/s41413-025-00469-2 (+) IF: 20.1 Q1	
2025	Yan et al., <i>Front Biosci</i>	Human CKD + mouse fibrosis; THBS4 knockdown reduces fibrosis	10.31083/FBL26076 (+) IF: 4.1 Q2	
2025	Cassiola et al., <i>PNAS</i>	THBS4 required for optimal CTL cytotoxic attack particles	10.1073/pnas.2413866122 (+) IF: 9.5 Q1	
2025	Tsujo et al., <i>Carcinogenesis</i>	CAF THBS4 promotes HER2 heterogeneity and trastuzumab resistance	10.1093/carcin/bgaf064 (+) IF: 3.8 Q2	
2024	Meng et al., <i>Pharmacol Res</i>	MLL4–THBS4 pathway protects cardiac ER homeostasis	10.1016/j.phrs.2024.107263 (+) IF: 12.2 Q1	

AI tools for literature search

<https://www.claude.com/>

Claude Chat

Lastest research especially with in 3 to 5 years and most recent research. Provide complete summary and in depth analysis of Thbs4 in health and diseases, specifically in aging, metabolic diseases and the pro and con of Thbs4 in health and diseases

Loaded tools, used Claude in Chrome integration

I tried to pull live literature (PubMed/browser) but no search tool or browser session is available in this conversation, so this synthesis is written from my own knowledge (reliable through ~May 2026). **I can't verify citations against a database, so treat every reference below as a pointer to check, not a confirmed citation.** I've flagged my confidence level throughout.

Let me write this up properly.

Created a file, read a file

A few things worth flagging about what you're getting:

On sourcing. I couldn't run PubMed, bioRxiv, or a browser session in this conversation, so this is written from my own knowledge (good through roughly May 2026) rather than from retrieved papers. I've tagged every claim as Established, Emerging, or Speculative, and \$12 gives you the exact queries to reconstruct and check it. Please verify the citations before using any of them — I can name papers but can't confirm the details.

The three points I'd argue are the most useful takeaways:

Write a message...



Opus 5 High



Claude Science

Lastest research especially with in 3 to 5 years and most recent research. Provide complete summary and in depth analysis of Thbs4 in health and diseases, specifically in aging, metabolic diseases and the pro and con of Thbs4 in health and diseases

I'll pull the actual literature rather than answer from memory. Let me start with the PubMed connector's method reference.

- ✓ Loaded 2 skills, ran a search, ran a command 4 steps · 1 failed
- ✓ Loading PubMed connector reference loaded
- ✓ Searching for literature review skills
- ✓ Loading literature review methodology skill
- ⌵ Searching PubMed for THBS4 literature

Background cell failed · 20m 0s

The connector call needed an approval that timed out. NCBI E-utilities is direct kernel instead — faster and batchable.

- ✓ Ran 4 commands
- ⌵ Fetching THBS4 records from NCBI E-utilities Module
- ✓ Fetching THBS4 records via urllib
- ✓ Downloading abstracts and parsing metadata
- ✓ Screening 450 abstracts for THBS4 relevance and topic

Aging and metabolic hits are sparse in the title/abstract sweep. Running target fill those axes.

- ✓ Ran 6 commands
- ✓ Running targeted aging and metabolic queries
- ✓ Querying Europe PMC full-text for THBS4 aging and metabolic papers
- ✓ Collecting open-access full-text candidates from Europe PMC
- ✓ Mining full texts for THBS4 mention contexts
- ⌵ Extracting THBS4 findings from mined full texts RuntimeError: host_call 'llm_batch' failed: host.llm() per...
- ✓ Ranking aging and metabolic full-text hits deterministically 4 lines of output

LLM budget for this session is spent, so I'll read the mined evidence directly. Pulling the aging and metabolic axes first.

- ✓ Printing aging-axis THBS4 evidence

DOI extraction picked up reference-list IDs — fixing that before I use any citation.

Notebook

Five gaps are specific enough to design against: no conditional *Thbs4* allele in any metabolic organ (all inference rests on global knockouts, which is unsafe given the cardiomyocyte/fibroblast inversion); the rejuvenation claim untested in vivo; tissue and plasma pools never measured together; MASLD essentially untouched despite THBS4 marking the pro-fibrotic fibroblast state in heart, kidney and lung; and a sex interaction that recurs across five independent findings but is almost never designed for.

- [THBS4_review_2026.md](#) — full review with mechanism, organ-by-organ ledger, therapeutic liabilities, and 133 inline citations across 112 DOIs
- [thbs4_evidence_table.csv](#) — the 344 screened papers with year, journal, DOI, topic tags, assigned direction of effect, species, and extracted finding, for your own filtering

One caveat on method: topic tagging and direction assignment in the CSV were model-assisted over abstracts, so the aggregate proportions in panel b are a literature-sentiment summary rather than a meta-analysis — every claim quoted in the review was checked back against the retrieved record, but the table's tags are best treated as a triage layer.

GENERATED · 3

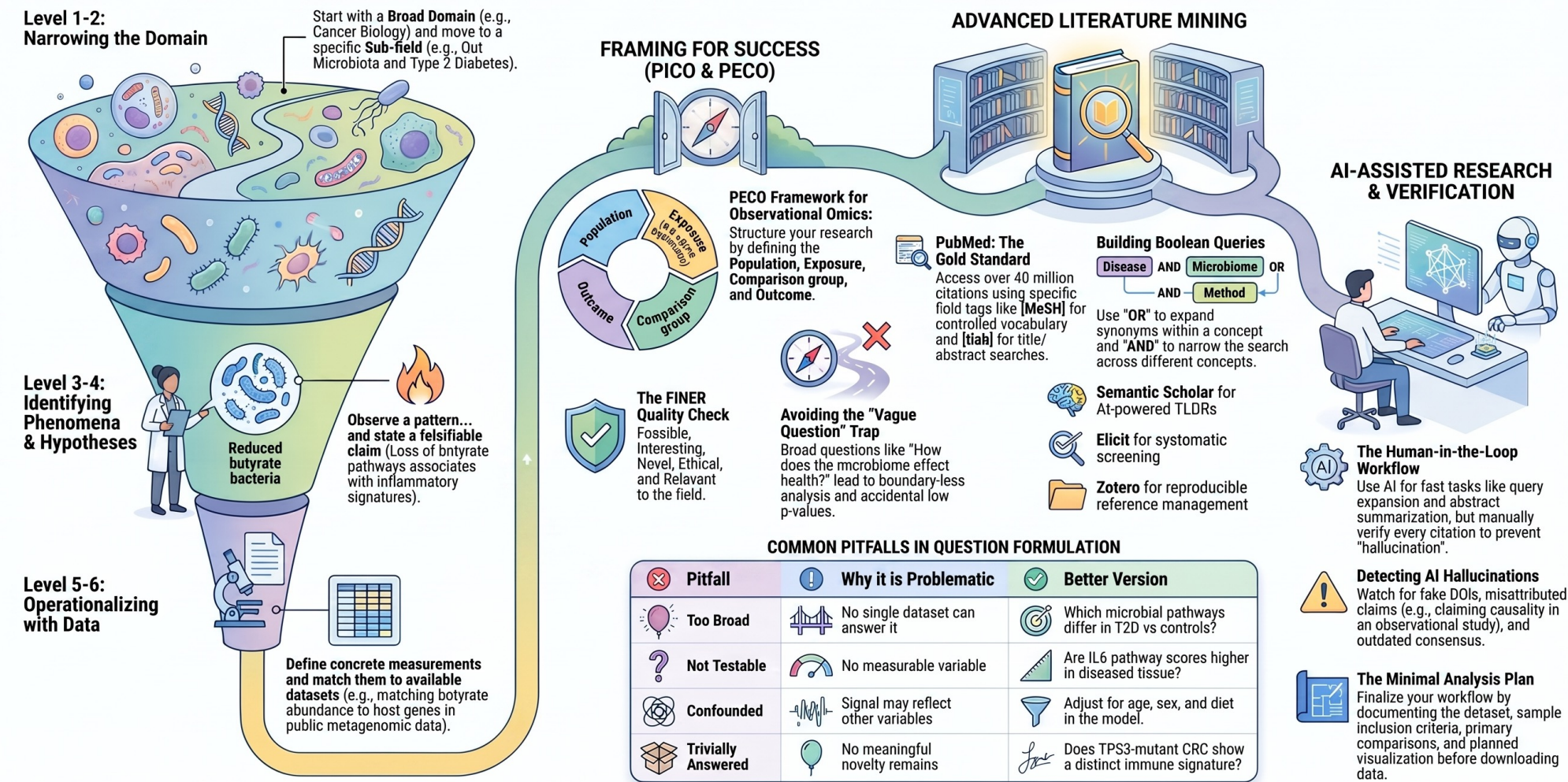
Thrombospondin-4 in disease
- Updated 2.0 for next 3 months (2026)
ThBS4..._ew_2026.md

thbs4_landscape.png

~340 rows · 12 columns
123 pmid
thbs4_e..._e_table.csv

9 minutes ago

The Bioinformatics Research Workflow: From Vague Curiosity to Data-Driven Evidence



We will have quiz next week.

Do not forget to review the class material.

5 multiple choice questions

1 min of each question for total 5 mins

AI accelerates search and analysis, but science decides what is true

Week 2 integrates four pillars of scientific reasoning

Question

Workflow

AI Acceleration

Verification

Pre-Reading for Week 3

Create an NCBI account | Review one GEO or SRA accession page

Ensure R and RStudio are installed | Bring your refined question + dataset

Week 2 Complete